MIL-S-23378D
22 January 1988
SUPERSEDING
MIL-S-23378C
24 October 1978

MILITARY SPECIFICATION

SCREW MACHINES, AUTOMATIC, SINGLE SPINDLE, 1/2 INCH THROUGH 2-3/8 INCHES SPINDLE CAPACITY

1. SCOPE

1.1 Scope. This specification covers single spindle, turret type screw machines with equipment and accessories specified herein.

1.2 Classification. The machines shall be of the following classes and sizes. The classes and sizes to be furnished shall be as specified (see 6.2.1).

Class I - 3 HP main drive motor (minimum). Maximum turning length not less than 1-1/2 inches.

Size	Maximum bar diameter (inches)
1/2	1/2
3/4	3/4

Class II - 7-1/2 HP main drive motor (minimum). Maximum turning length not less than 3-1/2 inches.

Size	Maximum bar diameter (inches)
3/4	3/4
1-1/4	1-1/4
1-5/8	1-5/8
2	2
2-3/8	2-3/8

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, ATTN: DIPEC-SSM, Memphis, Tennessee 38114-5051, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 3416

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T16:35Z
Check the source to verify that this is the current version before use.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issue of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-M-18058 - Machinery, Metal and Woodworking, Packaging of.

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services.

FED-STD-376 - Preferred Metric Units for General Use by the Federal Government.

MILITARY

MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interferences.

MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement of.

2.1.2 Other Government documents and publications. The following other Government document forms a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

U. S. DEPARTMENT OF LABOR

29 CFR 1910 - Occupational Safety and Health Standards (OSHA).

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402-0001.)

(Copies of specifications, standards, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issue of the documents which are DoD-adopted shall be those listed in the issue of the

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T16:35Z
Check the source to verify that this is the current version before use.

DoDISS specified in the solicitation. Unless otherwise specified, the issue of documents not listed in the DoDISS shall be the issue of the non-Government documents which are current on the date of the solicitation.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/NFPA 79 - Electrical Standard for Industrial Machinery.

ANSI B5.9 - Spindle Noses for Toolroom Lathes, Engine Lathes, Turret Lathes and Automatic Lathes (ISO 702).

ANSI B5.10 - Machine Tapers.

ANSI B11.13 - Single- and Multiple-Spindle Automatic Screw/Bar and Chucking Machines - Safety Requirements for the Construction, Care, and Use.

ANSI/ISO 54 - Cylindrical Gears For General Engineering and For Heavy Engineering, Modules and Diametrical Pitches of.

ANSI/NEMA MG 1 - Motors and Generators.

(Application for copies should be addressed to the American National Standards Institute, ATTN: Sales Dept., 1430 Broadway, New York, NY 10018-3363.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 3951 - Standard Practice for Commercial Packaging.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

AMERICAN GEAR MANUFACTURERS' ASSOCIATION (AGMA)

AGMA 360.02 - Manual for Machine Tool Gearing.

AGMA 390.03 - Gear Classification, Materials and Measuring Methods For Unassembled Gears.

(Application for copies should be addressed to the American Gear Manufacturers' Association, Standards Department, 1500 King Street, Suite 201, Alexandria, VA 22314-2717.)

(Non-Government standards and other publications are normally available from the organizations which prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T16:35Z
Check the source to verify that this is the current version before use.

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order (see 6.2.1), a sample shall be subjected to first article inspection (see 4.4 and 6.3).

3.2 Design. The machine shall be new (not a prototype) and one of the manufacturer's current models capable of operations in accordance with the requirements herein. The machine shall include all components, parts, and features necessary to meet the performance requirements specified herein. All parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement, and repair.

3.2.1 Measurement systems. Unless otherwise specified, either the U.S. Customary System of Units (US) or the International System of Units (SI) shall be used in the design and construction of the screw machine. When only one system of measurement is acceptable, the particular system required shall be as specified (see 6.2.1). In this specification, all measurements, dimensions, sizes, and capacities are given in US units. These measurements may be converted to SI units through the use of the conversion factors and methods specified in FED-STD-376.

3.2.1.1 Measuring and indicating device calibrations. Unless otherwise specified, either the U.S. customary System of Units (US) or the International System of Units (SI) shall be used to graduate measuring and indicating devices such as scales, depth stops, carriage stops, dial indicators, pressure gauges, temperature indicators, and other similar devices. When only one system of graduation is acceptable, the particular graduation required shall be as specified (see 6.2.1). Regardless of the measurement system used, all measuring and indicating devices on the screw machine shall be graduated in the same system. All feed dials shall have independent zero adjustments and shall be calibrated in such a manner that the last dial graduation progresses into and is continuous with the first dial graduation as the dial is rotated through the zero position.

3.2.2 Reclaimed materials. The machine may contain reclaimed materials provided such materials will not jeopardize the machine's intended use, performance, or design life. The materials shall have been reclaimed in a manner which will restore them to the same chemical composition and physical properties as the materials originally selected for use on the machine.

3.2.3 Energy efficiency. The machine and its components that directly consume energy in normal operation shall be designed and constructed for the maximum degree of energy efficiency as governed by the developments available within the industry.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T16:35Z
Check the source to verify that this is the current version before use.

3.2.4 Controls. All operating controls shall be located convenient to the operator at his normal work station.

3.2.5 Safety and health requirements. Covers, guards, or other safety devices shall be provided for the point of operation and all parts of the machine that present safety hazards. The safety devices shall not interfere with the operation of the machine. The safety devices shall prevent unintentional contact with the guarded part, and shall be removable to facilitate inspection, maintenance, and repair of the parts. All machine parts, components, mechanisms, and assemblies furnished on the lathe, whether or not specifically required herein, shall comply with all of the requirements of OSHA 29 CFR 1910 and ANSI B11.13. If a conflict arises between 29 CFR 1910 and ANSI B11.13, 29 CFR 1910 shall apply. Exceptions and additional requirements for safety and health shall be as specified (see 6.2.1).

3.2.6 Mercury restriction. The machine shall not contain mercury or mercury compounds nor be exposed to free mercury during manufacture.

3.2.7 Asbestos restriction. Asbestos and materials containing asbestos shall not be used on or in the machine.

3.2.8 Environmental protection. Under the operating, service, transportation, and storage conditions described herein, the machine shall not emit materials hazardous to the ecological system as prescribed by Federal, state, or local statutes in effect at the point of installation.

3.2.9 Lubrication. Means shall be provided to ensure adequate lubrication for all moving parts. Recirculating lubrication systems shall include a filter which is cleanable or replaceable. Each lubricant reservoir shall have at least a 24 hour capacity and means for determining fluid level. Means shall be provided to indicate a low oil condition. All oil holes, grease fittings, and filter caps shall be accessible.

3.2.10 Interchangeability. All parts subject to replacement shall be manufactured to definite dimensions and tolerances. Replacement parts (except for hand-fitted parts such as ways and boxed bearings) shall be interchangeable without requiring modification. Replacement parts are those which are subject to wear or failure such as gears, bearings, shafts, feed screws, drive belts, pumps, and seals. Electrical replacement parts shall include, but not be limited to, motors, relays, fuses, switches, magnetics, pushbuttons, light bulbs, braking mechanisms, and solid-state devices. Tooling and accessories normally supplied as standard items shall also be considered replacement parts. Foreign-sourced replacement parts shall be interchangeable with domestically supplied components or be logistically supportable and readily available from a domestic source.

3.3 Construction. The machine shall be constructed of parts which are new, without defects, and free of repairs. The structure shall withstand all forces

Source: https://assist.dla.mil Downloaded: 2026-10-19T16:35Z
Check the source to verify that this is the current version before use.

encountered during operation of the machine to its maximum rating and capacity without distortion.

3.3.1 Castings and forgings. All castings and forgings shall be free of defects, scale and mismatching. No processes such as welding, peening, plugging, or filling with solder or paste shall be used for reclaiming any defective part.

3.3.2 Fastening devices. All screws, pins, bolts, and other fasteners shall be installed to prevent change of tightness. Fastening devices subject to removal or adjustment shall not be swagged, peened, staked, or otherwise permanently installed.

3.3.3 Surfaces. All surfaces shall be clean and free of harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

3.3.4 Welding, brazing, or soldering. Welding, brazing, or soldering shall be employed only where specified in the original design. None of these operations shall be employed as a repair measure for any defective part.

3.3.5 Painting. Unless otherwise specified (see 6.2.1), the machine shall be painted in accordance with the manufacturer's commercial practice.

3.3.6 Threads. All threaded parts used on the machine and its related attachments and accessories shall conform to FED-STD-H28 and the applicable "Detailed Standard" section referenced therein.

3.3.7 Dials. Rotating dials or handwheels shall be used for longitudinal movement or cross motion and shall be graduated in increments not greater than 0.10 inch for longitudinal and 0.001 inch for cross motion. Such graduations shall accurately indicate machine component movement or material removal. Dial and handwheel circumferences shall be permanently and legibly engraved or etched on a nonglare background and be of such size that graduations can be read from the operator's normal position. Feed dials shall have independent zero adjustments and be calibrated in such a manner that the last dial graduation progresses into and is continuous with the first dial graduation as the dial is rotated through the zero position.

3.3.8 Gears. All gears, shafts, and pinions of spindle and axis drive train shall be designed and manufactured of a suitable ferrous material to meet or exceed the requirements of AGMA 360.02 and 390.03 for the English (US) system or ISO 54 for the metric (SI) system. The gears shall be of proper width and size to transmit full-rated torque and horsepower throughout the speed ranges without failure for the expected service life of the machine. Gears, shafts, and pinions in the drive train shall be hardened and ground steel. Hardness shall be not less than Rockwell C-45.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T16:35Z
Check the source to verify that this is the current version before use.

3.3.9 Electromagnetic interference control. When specified (see 6.2.1), equipment furnished under this specification shall comply with MIL-STD-461. The equipment and subsystems class and the emission and susceptibility requirements shall be as specified.

3.4 Components. The machine shall consist of not less than a bed and base, spindle and spindle drive, stock feeding mechanism, indexing tool turret, cross slide and upper rear slide, and operating mechanisms.

3.4.1 Bed and base. The machine bed shall protect the slide mechanisms from chips, dirt, and coolant. The chip compartment shall have a perforated divider to separate chips from coolant being returned to the reservoir. Slide ways, for supporting and guiding the indexing turret slide and cross slides, shall have replaceable steel ways having a surface hardness not less than Rockwell C-60. A removable tray shall be mounted in the bed for catching completed work parts. The base of the machine shall be equipped with means for securing and leveling the machine on a foundation.

3.4.2 Spindle and spindle drive. The spindle shall be designed for advancing bar stock through the spindle. The spindle nose shall be designed for readily changing collets and in accordance with the requirements of ANSI B5.9. The spindle drive mechanism shall rotate the spindle in both directions by a multiple chain and gear drive. The spindle internal taper shall conform to the requirements of ANSI B5.10. Means for adjusting chain tension shall be provided. Speed changes shall be made automatically during the machining cycle. Spindle speeds and speed ratios shall be obtained by changing gears to provide the required speeds within the range of the machine. The spindle shall be mounted in preloaded ball or roller bearings. All rotating shafts in the drive mechanism shall be mounted in ball or roller bearings. A metal plate shall be permanently mounted convenient to the gear box showing the gear combinations for obtaining the available speeds.

3.4.3 Stock feeding mechanism. The stock feeding mechanism shall automatically advance bar stock through the spindle to a swing stop. The swing stop shall be adjustable for part length and timing to operate with the stock feed mechanism. The bar shall be advanced any length in one movement with the capacity of the machine. The feed mechanism shall have means for gaging the stock feed stroke and adjusting the grip on the bar diameter. The feed mechanism drive sleeve speed shall be independent of the main drive shaft speed and shall be provided with speed change gears. An electrical stopping device shall stop the machining cycle and give a warning signal when a bar of stock is expended.

3.4.4 Indexing tool turret. The standard indexing tool turret shall have six index positions with provision for double indexing to alternate positions. The tool turret shall be mounted on the turret slide in ball or roller bearings. The turrets shall be designed for mounting interchangeable round shank tool holders. The turret slide shall provide feed movement by cam action. Provisions shall be made to compensate for wear on the turret slide ways. The turret slide stroke

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T16:35Z
Check the source to verify that this is the current version before use.

shall have a longitudinal screw adjustment to suit various cam lobe heights and tooling. The turret slide shall be equipped with a rapid pull-out-and-return mechanism for chip clearance during deep drilling operations. Each class II machine shall have a positive stop to limit forward motion of the slide.

3.4.5 Cross slides and upper rear slide. Cross slides shall be mounted on the horizontal bed ways at both the front and rear of the spindle, each providing tool feeds at a right angle to the spindle axis. Each slide shall be cam operated. An adjustment screw with a dial graduated in 0.001 inch increments shall be provided for transverse adjustment on each cross slide. Cross slides shall have standard size T-slots for mounting interchangeable tool holders. The vertical slide shall be used for clamping various size cut-off blades. Each cross slide and the vertical slide shall operate independently and in combinations. Each class II machine shall have a micrometer adjustment to set the cut-off slide.

3.4.6 Operating mechanisms. Cam holders for mounting the slide operating cams shall be graduated and shall be adjustable for angular position. Mechanical devices for automatically actuating machine functions shall be adjustable for positioning, or shall be graduated drum type carriers. Automatic functions shall be performed in any required sequence of operation including: turret index, stock feed, spindle speed change and spindle reverse, work deflector, and auxiliary functions, as provided. Manually operated electrical controls shall include spindle start and stop, drive start and stop, and jog functions. The main drive shaft shall operate at a constant speed as established by change gears. A handwheel shall be provided to cycle the machine during set-up. A safety interlock shall prevent the drive shaft from operating unless the handwheel is disengaged. Each slide shall be provided with means for hand operation during set-up. The drive shaft shall be operated by a separate electric motor.

3.4.7 Change gears. Gears shall be changeable without any machine disassembly. The gears shall be locked on the gear shafts with means to prevent gears working loose during operation. A component shall be provided in the machine base for storing gears not in use.

3.4.8 Coolant system. The coolant system shall include a sump or reservoir, a power driven pump and all necessary piping. The sump or reservoir shall have sufficient capacity to permit full flow of coolant. The system shall have means for draining and cleaning and shall include a baffle or a strainer which is easily removable for cleaning. Means shall be provided to permit the operator to direct and control the amount of coolant.

3.4.9 Electrical system. The electrical system shall conform to ANSI/NFPA 79. Unless otherwise specified (see 6.2.1), each machine shall draw all of its electrical power from a single 230/460-volt, 3-phase, 60-hertz (Hz) circuit. Each machine shall have a fused safety disconnect switch or circuit breaker and be initially wired for operation on 460 volts.

3.4.9.1 Motors. Motors shall be rated for continuous duty and shall have ball or roller bearings of the sealed, balanced, and permanently lubricated type. Alternating-current (AC) motors shall be designed to operate on 60-Hz. Unless

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T16:35Z
Check the source to verify that this is the current version before use.

otherwise specified (see 6.2.1), each motor enclosure shall meet the requirements for a drip-proof enclosure. All motors shall meet the requirements of ANSI/NEMA MG 1 and ANSI/NFPA 79. Each motor shall have an identification plate to identify manufacturer, model, serial number, voltage, amperage, horsepower, phase, and frequency.

3.4.9.2 Control circuit voltage. Auxiliary control circuits shall be isolated from the input electric power supply by a low-voltage transformer having a secondary nominal voltage of 120 volts or less.

3.5 Size and capacity. Unless otherwise specified (see 6.2.1), the size and capacity of the machine shall be not less than those specified in table I for the machine class and size specified.

3.6 Performance. The machine and components, operator's controls, and safety devices shall function properly when tested in accordance with 4.8.1 and 4.8.3.

3.6.1 Additional performance requirements. Performance features required in addition to, or in lieu of, the requirements herein shall be as specified and fully described (see 6.2.1).

3.7 Alignment tolerances. Each machine shall meet the alignment tolerances specified in table II when tested according to 4.8.2.

3.8 Worklight. Unless otherwise specified (see 6.2.1), a worklight shall be provided. The light shall have a protective shield, an on-off switch, and shall be attached to provide for adjustment to reflect maximum light on the work area. If a built-in lighting system is provided, the adjustable light described above shall not be required.

3.9 Standard equipment. Unless otherwise specified (see 6.2.1), the following equipment shall be supplied with each machine.

3.9.1 Stock stands and tube. The stock stands shall support the stock tube and additional bars. Provisions shall be made in the stand bases for securing to the foundation. The stock tube shall suppress noise produced by the rotating bar and fully enclose a 12 foot long bar when it is inserted into the machine.

3.9.2 Change gears. A complete set of change gears shall be furnished to provide all spindle speeds and cycle speeds within the range of the machine.

3.9.3 Collets and feed fingers. A complete set of spindle collets and feed tube fingers shall be furnished as required, (see 6.2.1), for gripping round, square, and hexagon bars of diameters within the capacity of the machine. (Unless otherwise specified under 6.2.1, round collets shall be in 1/32 inch increments; square and hexagon collets shall be in 1/16 inch increments.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T16:35Z
Check the source to verify that this is the current version before use.

3.9.4 Turret drilling attachment. The drill attachment shall be a complete unit, consisting essentially of the drilling spindle for mounting in the tool turret, an electric drive motor, and drive mechanism. Maximum drill capacity shall be at least 3/32 inch diameter for class I machines and 3/16 inch diameter for class II machines.

3.9.5 Cam shaft accelerator (class II machines). The cam shaft accelerator shall automatically accelerate the cam shaft to reduce noncutting time when indexing tools. The accelerator mechanism shall be electrically controlled, actuated by adjustable trips.

3.9.6 Longitudinal turning slide. The turning slide shall be a complete unit for mounting on the front cross slide. It shall be used for turning straight and taper diameters within the bar capacity of the machine. Turning length shall be at least 1-3/8 inches for class I machines and 3 inches for class II machines. The tool holder shall accept a square tool at least 5/16 inch in size on class I machines and 1/2 inch in size on class II machines.

3.9.7 Single point threading attachment for class II machines. The threading attachment shall be a complete unit, consisting essentially of a tool holder mounted on the front cross slide having longitudinal movement, and shall be synchronized with and driven from the machine spindle. The attachment shall thread diameters up to the bar capacity of the machine, and at least 2 inches in length. The attachment shall be adjustable to at least 7 degrees in both directions for taper threading.

3.9.8 Turret tool coolant system for class II machines. The coolant system shall be a complete unit for delivering cutting coolant through any tool mounted in the turret, when indexed to the cutting position.

3.9.9 Double grip collets (class II machines). When specified, (see 6.2.1), double grip collet sets (collets having two gripping surfaces contacting the stock the length of the collet grip surfaces) shall be provided. Each collet shall be designed to hold stock sizes in fractional inch, decimal, and metric sizes and cover the stock size capacity of the machine.

3.10 Optional tooling. When specified, the following optional equipment shall be provided (see 6.2.1). For class I machines (1/2 to 3/4 inch), use 5/8-inch diameter tools. For class II machines (3/4 to 2-3/8 inches), use 1-inch diameter tools.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T16:35Z
Check the source to verify that this is the current version before use.

a. Box tools
b. Slide tools
c. Drilling and chamfering
d. Adjustable holders
e. Drilling and turning
f. Reamer holders
g. Collet holders
h. Thread roll holders
i. Tap Holders
j. "V" tool holders
k. Boring tools and holders
l. Tool post
m. Knee tools
n. Centering and facing
o. Pointing tools
p. Trepanning tools
q. Knurl holders
r. Swing tools
s. Guides
t. Recessing tools
u. Back rests
v. Turret tool adapters
w. Bushings
x. Hollow mills
y. Bushing tools
z. Stock stops

3.11 Repair parts. Required repair parts shall be furnished as specified (see 6.2.1). All repair parts shall be assigned a part number identified in a repair parts manual.

3.12 Marking on plates. All words on plates shall be in the English language. Characters shall be engraved, etched, embossed, or stamped in boldface on a contrasting background. All plates shall be corrosion-resistant.

3.12.1 Lubrication plate. Unless otherwise specified (see 6.2.1), a lubrication plate shall be permanently and securely attached to each machine. The plate shall contain the following information:

Points of lubricant application
Servicing interval
Type of lubricant
Viscosity

3.12.2 Nameplate. A nameplate shall be securely attached to each machine. Unless otherwise specified (see 6.2.1), the nameplate shall contain the information listed below. If the machine is a special model, the model designation shall include the model number of the basic standard machine and a suffix identified in the manufacturer's permanent records.

Nomenclature
Manufacturer's name
Manufacturer's model designation
Manufacturer's serial number
Power input (volts, total amps, phase, frequency)
Contract Number or Order Number
National Stock Number or Plant Equipment Code
Date of manufacture

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T16:35Z
Check the source to verify that this is the current version before use.

3.13 Hour meter. When specified (see 6.2.1), the machine shall be equipped with an hour meter. The hour meter shall be installed to display accumulated operating time of the main drive motor. The meter shall be of the nonresetting type and shall have a range of 0 to 99,999 hours in increments of not greater than 1 hour. The meter shall be sealed to prevent the entrance of dust and moisture and shall be mounted to withstand shock and vibration generated by the machine.

3.14 Technical data. When technical data is required, it shall be furnished in accordance with the requirements of DD Form 1423. All technical data furnished shall be written in the English language.

3.15 Workmanship. Workmanship of the machine shall be commensurate with all requirements specified herein and shall be of a quality equal to that prevailing among manufacturers producing equipment of the type covered by this specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

c. Acceptance test (see 4.6).

4.3 Inspection conditions. Unless otherwise specified (see 6.2.1), all inspections, tests, and examinations shall be performed in an indoor facility with ambient conditions of temperature: 55°F to 85°F; humidity: 10 percent to 90 percent relative humidity; and atmospheric pressure: 15.0 to 13.0 psi.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T16:35Z
Check the source to verify that this is the current version before use.

4.4 First article inspection. When a first article inspection is required, it shall be applied to the first article submitted in accordance with 3.1. Unless otherwise specified (see 6.2.1), first article inspection shall consist of the examination in 4.7 and all test in 4.8. Failure of the item to pass the first article examination or any test shall be cause for rejection.

4.5 Quality conformance inspection. Quality conformance inspection shall be applied to each item prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2.1), quality conformance inspection shall consist of the examination in 4.7, the tests in 4.8, and the inspection in 4.9. Failure of the item to pass the examination, the tests, or the inspection shall be cause for rejection.

4.6 Acceptance test. Acceptance test shall be performed on the machine to ensure conformance with this specification. Unless otherwise specified (see 6.2.1), the acceptance test shall be performed only after the machine is installed at its final location. The acceptance test shall consist of the examination in 4.7 and all tests in 4.8. Failure of the machine to pass one or more tests or any part of a test will be cause for rejection.

4.7 Examination. The machine shall be examined to determine compliance with all requirements of this specification including compliance with all applicable standards cited in Section 2.

4.8 Tests.

4.8.1 Operational test. The machine shall be operated at no load for not less than 60 minutes. Proper operation of all controls, motors, adjusting mechanisms, and accessories shall be verified during the trial period.

4.8.2 Alignment tolerance test. Each machine shall be tested to determine that it complies with the alignment tolerance requirements of 3.7.

4.8.3 Performance tests. The screw machine shall be subjected to the following turning tests to determine conformance to the accuracies of table III. At the option of the procuring agency, each machine shall be subjected to tests in accordance with specific requirements designated by the activity in lieu of or in addition to the following tests (see 6.2.1).

4.8.3.1 Turret turning test. A bar of free machine steel of suitable diameter within the machine spindle capacity shall be held in the machine spring collet and turned with a single point tool held in a turret tool holder with a blade position of 3 or 9 o'clock for horizontal test and 6 or 12 o'clock for vertical tests.

4.8.3.2 Cross slide forming test. Material as specified in 4.8.3.1 shall be plunge-formed for a 1-inch length to a suitable diameter using a circular, flat, or dovetail form tool from each of the machine's horizontal cross slides.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T16:35Z
Check the source to verify that this is the current version before use.

4.8.4 Electromagnetic interference control tests. Equipment requiring electromagnetic interference control shall be tested for compliance with 3.3.9 using the test procedures given in MIL-STD-462.

4.9 Packaging inspection. Packaging of each item shall be inspected to determine compliance with the requirements of section 5.

5. PACKAGING

5.1 Preservation, packing, and marking. Unless otherwise specified, preservation, packaging, and marking shall be in accordance with ASTM D 3951. When specified (see 6.2.1), level A preservation, level A or level A or level B packing, and marking shall be accomplished in accordance with MIL-M-18058.

6. NOTES

6.1 Intended use. The automatic screw machines covered by this specification are intended for use in any machine shop for producing a variety of products from bar stock. Machining operations include automatic turning, drilling, forming, threading, and slotting. The machines covered by this specification are of one type. The basic difference between classes is that class II machines have more work length capacity, more horsepower, and more possible attachments than are available for class I machines.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Class and size required (see 1.2).

c. First article, if required (see 3.1).

d. If machine is required to be configured in a specific measurement system (US or SI), stated required system (see 3.2.1).

e. If measuring and indicating devices are required to be graduated in a specific measurement system (US or SI), stated required system (see 3.2.1.1).

f. Additional safety and health requirements, if required (see 3.2.5).

g. Painting, if different (see 3.3.5).

h. If electromagnetic interference control is required, specify the equipment and subsystem class and the emission and susceptibility required (see 3.3.9).

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T16:35Z
Check the source to verify that this is the current version before use.

i. Electrical system and input power, if different (see 3.4.9).

j. Motor enclosure, if different (see 3.4.9.1).

k. Size and capacity, if different (see 3.5).

l. Additional performance features, if required (see 3.6.1).

m. Worklight, if different (see 3.8).

n. Standard equipment, if different (see 3.9).

o. Collets and feed fingers, as required (see 3.9.3).

p. Double grip collets shall be provided when specified (see 3.9.9).

q. Specify and fully describe optional equipment, if required (see 3.10).

r. Specify applicable repair parts, if required (see 3.11).

s. Lubrication plate, if different (see 3.12.1).

t. Nameplate, if different (see 3.12.2).

u. Hour meter, if required (see 3.13).

v. Inspection conditions, if different (see 4.3).

w. First article inspection, if different (see 4.4).

x. Quality conformance inspection, if different (see 4.5).

y. Acceptance test, if different (see 4.6).

z. Performance test, if different (see 4.8.3).

aa. If preservation, packing, and marking in accordance with MIL-M-18058 is required, specify level required and specification number (see 5.1).

6.2.2 Contract data requirements. Required technical data (operating manuals, parts, lists, wiring diagrams, foundation and anchor bolt plans, and acceptance test reports) should be specified on a DD Form 1423, Contract Data Requirements List, incorporated into the contract.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T16:35Z
Check the source to verify that this is the current version before use.

6.3 First article. When first article inspection is required, the item to be tested should be the first item offered for acceptance under the contract. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

6.4 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

6.5 Inspection facilities. The contracting officer should be notified by each contractor in his response to the solicitation of the location where inspection tests are to be performed. It is recommended all inspections be performed within the limits of the continental United States.

6.6 Subject term (keyword) listing:

Automatic
Screw machine
Turret type

6.7 Training. Training required by the procuring activity shall be provided as specified in the contract.

6.8 Warranty. Warranty requirements shall be as specified in the contract by the procuring activity. The warranty period shall commence upon completion of the installation and final acceptance test of the machine.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T16:35Z
Check the source to verify that this is the current version before use.

TABLE 1. Sizes and capacities.

MACHINE CHARACTERISTIC	REQUIREMENT (Dimensions in Inches)						
	Class 1		Class 2				
	1/2 inch	3/4 inch	3/4 inch	1 1/4 inch	1 5/8 inch	2 inch	2 3/8 inch
Collet capacity - maximum diameter round bar, inches	1/2	3/4	3/4	1 1/4	1 5/8	2	2 3/8
Turning length - maximum, inches	1 1/2	1 1/2	3 1/2	3 1/2	3 1/2	3 1/2	3 1/2
Spindle speeds:							
Total range, RPM - 2 speed drive	34 to 8665	34 to 6060	20 to 5018	14 to 3500	14 to 2900	14 to 2484	14 to 2484
Number of speeds - High speed	18	17	18	18	17	17	17
Second speed	13	13	13	13	13	13	13
Spindle motor horsepower	3	3	7 1/2	7 1/2	7 1/2	7 1/2	7 1/2
Optional spindle speeds:							
Total range, RPM - 4 speed drive	6 to 8665	6 to 6060	4 to 5018	3 to 3500	3 to 2900	3 to 2484	3 to 2484
Number of speeds - High speed	18	17	18	18	17	17	17
Second speed	11	11	10	10	10	10	10
Third speed	13	13	13	13	13	13	13
Fourth speed	165	165	130	130	130	130	130
Spindle motor horsepower with optional speeds	7 1/2	7 1/2	7 1/2	7 1/2	7 1/2	7 1/2	7 1/2
Drive shaft speeds, RPM	120/240	120/240	120	120	120	120	120
Drive sleeve speeds, RPM	120/240	120/240	120/240	120/240	120/240	120/240	120/240
Bar feed in inches, single movement,							
drive sleeve at 120 RPM	2	2	4	4	4	4	4
drive sleeve at 240 RPM	1	1	2 3/4	2 1/2	2 1/2	2 1/4	2 1/4
Cycle time in seconds:							
Drive shaft at 120 RPM	1 1/2 to 100	1 1/2 to 100	1 2/3 to 1000	1 2/3 to 1000	1 2/3 to 1000	1 2/3 to 1000	1 2/3 to 1000
Drive shaft at 240 RPM	3/4 to 50	3/4 to 50	-	-	-	-	-
Tool turret - number of index positions	6	6	6	6	6	6	6
Diameter of turret tool holes, inches	5/8	5/8	1	1	1	1 1/4	1 1/4
Longitudinal adjustment of turret slide, inches	1/2	1/2	1	1	1	1	1

NOTE: Machine size and capacities shall be not less than the stated requirements of the size ordered nor greater than the requirements of the next larger size shown in above table. When the largest size and capacity machine is ordered the size and capacity of the machine offered shall not exceed the stated requirements by more than 10 percent. When a range is shown, the required performance is from the minimum or less to the maximum or greater.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T16:35Z
Check the source to verify that this is the current version before use.

Table II. Alignment tolerances.

Test	Requirements	
	Class 1	Class 2
Turret slide ways parallel to spindle		
Horizontal check, spindle end forward	0 to 0.0005" in 12"	0 to 0005" in 18"
Vertical check, spindle end high	0 to 0.0015" in 12"	0 to 0.0015" in 18"
Alignment of cross slide ways with spindle (sweep front and back ways)	0.006" in 18" sweep	0.008" in 24" sweep
Turret tool holes aligned with spindle		
Horizontal check, toward rear	0.0007"	0.0007"
toward front	0.0007"	0.0007"
Vertical check, high	0.0007"	0.0007"
low	0.0007"	0.0007"

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T16:35Z
Check the source to verify that this is the current version before use.

Table III. Performance accuracies.

Turret turning (ways parallel to machine spindle)	
Horizontally	Straight to
Vertically	.00025 per inch
Cross slide forming (repeatability and rigidity test)	
Diameter tolerance ±0.0005 inch	
Surface finish 125 microinch with no visible chatter	

Custodians:
Army - AL
Navy - SH
Air Force - 99

Review activities:
Army - AL
Navy - SH
Air Force - 84

User activities:
Navy - AS, MC

Preparing activity:
DLA - IP

Project (3416-0211)

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T16:35Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEFENSE LOGISTICS AGENCY
Headquarters
Defense Industrial Plant Equipment Center
Memphis, Tennessee 38114-5051

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 4966 WASHINGTON D. C.
POSTAGE WILL BE PAID BY THE DEFENSE LOGISTICS AGENCY

Commander
Defense Industrial Plant Equipment Center
ATTN: DIPEC-SSM
Memphis, Tennessee 38114-5051

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T16:35Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-S-23378D	Screw Machines, Automatic, Single Spindle, 1/2 Inch Through 2-3/8 Inches Spindle Capacity

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

Source: https://assist.dla.mil – Downloaded: 2026-10-19T16:35Z
Check the source to verify that this is the current version before use.